INFORMATICS INSTITUTE OF TECHNOLOGY

3D INTERACTIVE MEDIA DEVELOPMENT

Design and Documentation Group Report

Coursework 1 (CW 1) | Academic Year 2026/27

Module Name	3D Interactive Media Development
Module Code	5MMCS007C.1 (University of Westminster module 5MMCS007W)
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Submission Deadline	12 November 2026
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1. Executive Summary of Individual Contributions

As a second-year Computer Science undergraduate leading key design and technical components for the Agile Team Project, my primary contributions to Coursework 1 (CW1) focused on 3D asset modeling for animal enclosures, engine UI canvas implementation, documentation management, and GitHub version control structure.

While initial concepts planned for real-time spatial audio and live animal AI state machines, our iterative sprint testing highlighted spatial collision and UI menu focus priorities. Consequently, my technical scope for CW1 was strictly tailored to solidifying environmental 3D models (Bear, Duck, and Horse enclosures), low-poly asset creation (Lego horses, cockatoos, and parrots), constructing the engine Main Menu & Pause Menu UI Canvases, authoring the full CW1 Group Report, and managing repository governance.

2. Detailed Breakdown of Completed Deliverables

2.1. 3D Environment Modeling & Asset Creation (Design)

Designed and constructed the structural 3D Diorama layouts for three primary terrestrial zones inspired by Dehiwala Zoo: the Bear Enclosure, Duck Enclosure, and Horse Enclosure. Applied custom PBR materials and box colliders across perimeter fences and natural boundaries to maintain physical affordance.

Created custom low-poly / Lego-style 3D models including stylized horses, cockatoos, and parrots. These assets were prepared and converted into Unity prefab formats, ready to be instantiated inside active scene triggers during CW2 implementation.

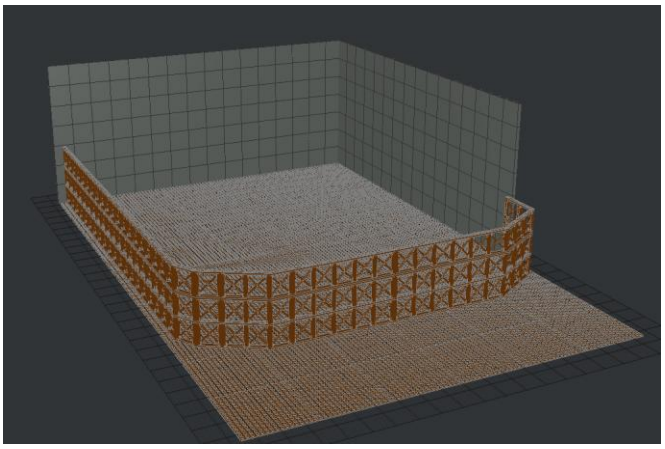


Figure 1: Horse enclosure

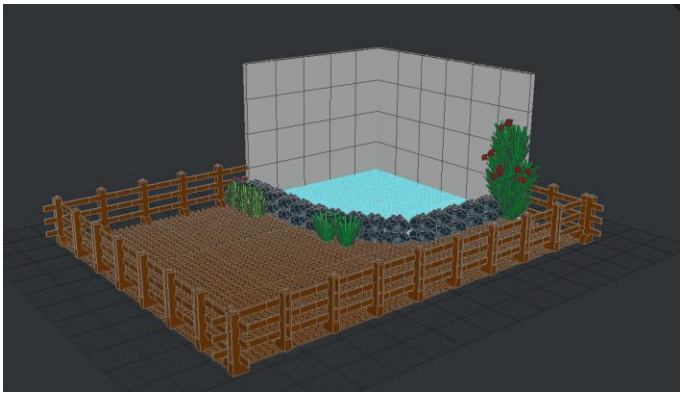


Figure 3: Duck enclosure

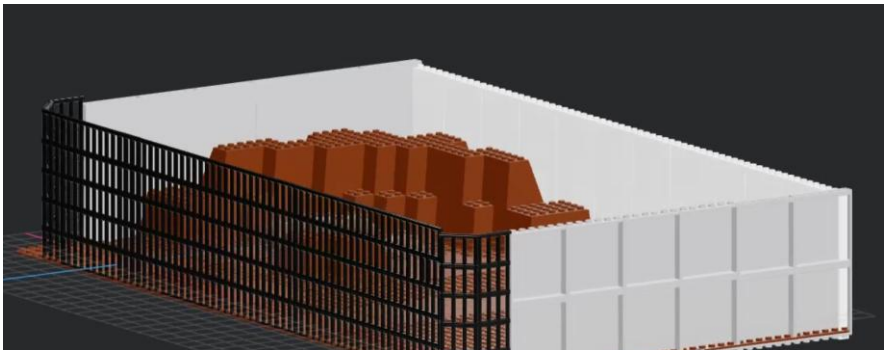


Figure 2: Bear enclosure

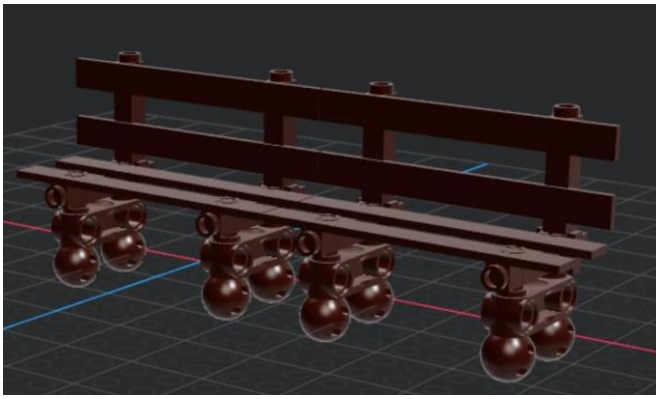


Figure 4: Bench

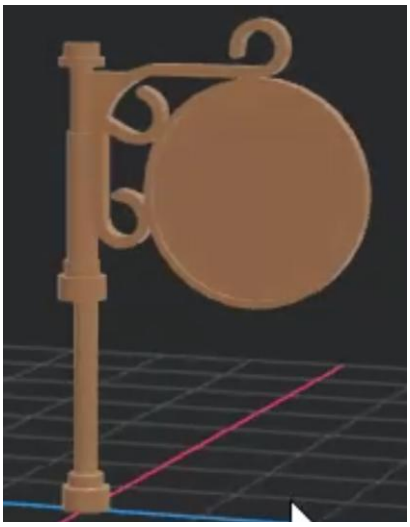


Figure 6: Signpost



Figure 5: Horse

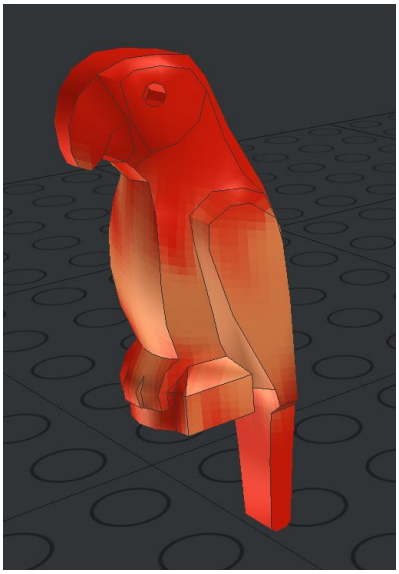


Figure 7: Parrot

2.2. Unity UI Canvas & Main Menu Implementation (Coding)

Designed and implemented the core Screen-Space Overlay UI Canvas prior to filming the group project demonstration video. Built interactive main menu controls comprising Play, Controls, Audio/Graphics Settings, and Exit buttons.

Scripted the C# Pause Menu toggle logic tied to the Escape key (`CursorLockMode.None` vs `CursorLockMode.Locked`). This enabled seamless switching between raw mouse-look camera navigation during FPV exploration and unlocked cursor pointer interaction when adjusting simulation options.

2.3. Project Documentation & Repository Management (Management)

Authored and compiled the comprehensive 10-section Group CW1 Report, ensuring strict alignment with academic 3DUI principles, FPV egocentric navigation theory, and Dehiwala Zoo layout references.

Initialized and governed the official group GitHub code repository. Structured version control branches, managed commit logs for research design files, WIP video logs, meeting minutes, and compiled scene assets.

3. Summary of AI Collaboration & Chat Transcripts

Throughout CW1, generative AI tools were utilized strictly as a thought partner and technical scribe to streamline project planning, refine documentation structures, and structure meeting minutes without replacing original code generation or 3D asset modeling.

Key Areas of Generative AI Assistance:

• **Planning & Outline Structuring:** Formulated the structural 10-point outline for the Group CW1 Report based on module guidelines.

• **Meeting Minutes Formatting:** Synthesized raw team notes into structured meeting minutes covering initial concept meetings, physical peer-review analysis at IIT campus, and final demo recordings.

• **3DUI Theory Refinement:** Evaluated control-to-task mapping (WASD + Mouse-look) and cognitive load trade-offs within First-Person egocentric navigation framework.

4. Critical Reflection on AI Code Assist & Best Practice

In accordance with academic integrity guidelines, GitHub Copilot and AI assistants were evaluated for responsible integration within C# scripting workflows.

AI code assistance proved highly effective for generating repetitive UI canvas event listeners and basic cursor state locks (`CursorLockMode`). However, initial AI-suggested locomotion snippets lacked frame-rate independence, requiring manual refactoring to enforce `Time.deltaTime` multipliers across movement vectors.

Furthermore, trigger volume logic was manually optimized to ensure boundary collision checks did not cause frame drops during UI canvas instantiations. Using AI for high-level scoping while strictly auditing and refactoring generated code ensured maintainability and deep technical understanding.

5. Planned Roadmap for Coursework 2

Building upon the solid architectural foundation established in CW1, my individual deliverables for CW2 will focus on bringing the modeled assets to life inside the active Unity scene:

CW2 Feature Domain	Planned Individual Contribution & Implementation
Animal Model Placement	Instantiate and place modeled horses, bears, ducks, cockatoos, and parrots directly into their respective enclosure zones.
Proximity Audio Triggers	Script C# `OnTriggerEnter` trigger volumes in front of enclosures to initiate 3D spatial animal calls upon player approach.
Informational Tag HUD	Develop 2D HUD pop-up panels displaying species biology facts when inspecting enclosure signage.
Animal AI & Animations	Implement basic Animator state machines for roaming walk cycles and idle behaviors.